Switching Circuits Laboratory (CS29002)

Assignment 4

February 04, 2026

About the Assignments

a) Build an 8-to-1 MUX using two 4-to-1 MUXs and basic gates. Hence realize a given Boolean function.

b) Realize a 3-way 4-bit data switch using 2-to-1 MUXs.

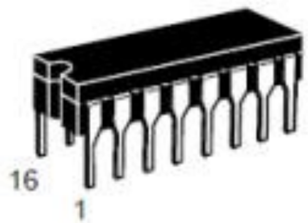
c) Realize a 4-bit binary to hexadecimal 7-segment decoder using an Erasable Programmable Read-Only Memory (EPROM).

For Part(a):

SELECT INPUTS		INPUTS (a or b)					OUTPUT
S_0	S_1	$\overline{E}$	I_0	I_1	I_2	I_3	Z
X	X	H	X	X	X	X	L
L	L	L	L	X	X	X	L
L	L	L	H	X	X	X	H
H	L	L	X	L	X	X	L
H	L	L	X	H	X	X	H
L	H	L	X	X	L	X	L
L	H	L	X	X	H	X	H
H	H	L	X	X	X	L	L
H	H	L	X	X	X	H	H

H = HIGH Voltage Level
L = LOW Voltage Level
X = Don't Care

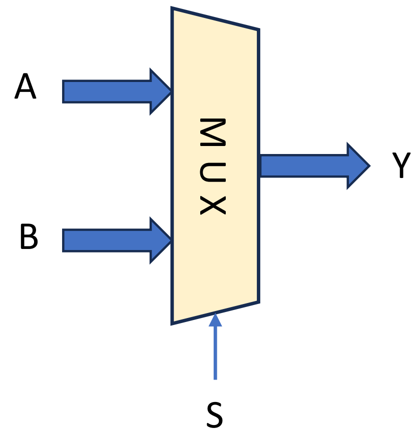
TRUTH TABLE

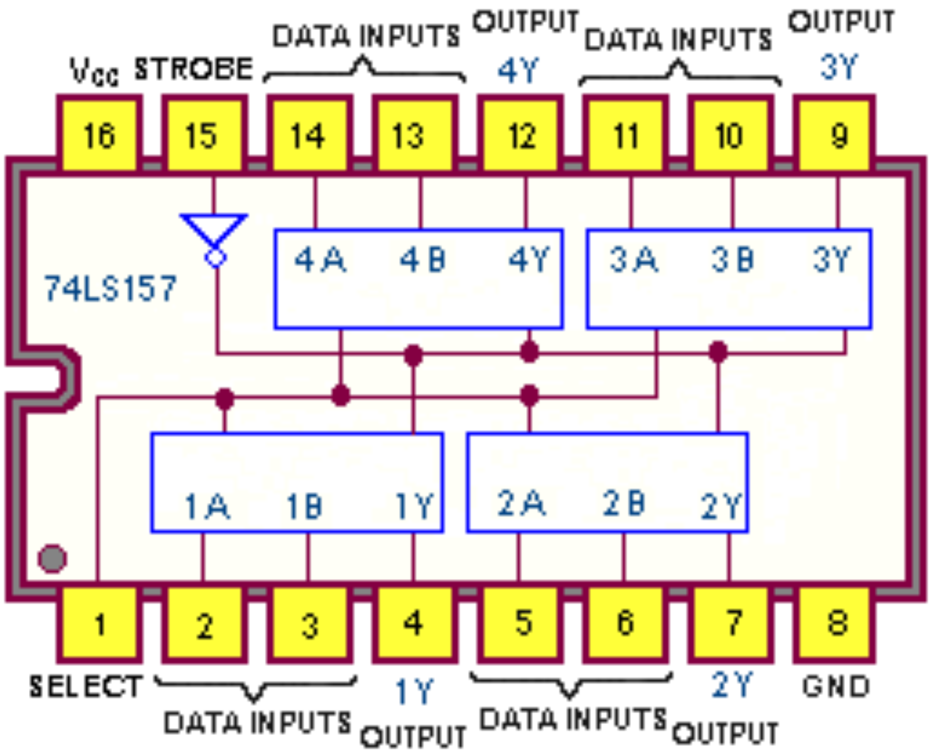


V_{CC} = PIN 16
GND = PIN 8

74LS153

For Part(b):

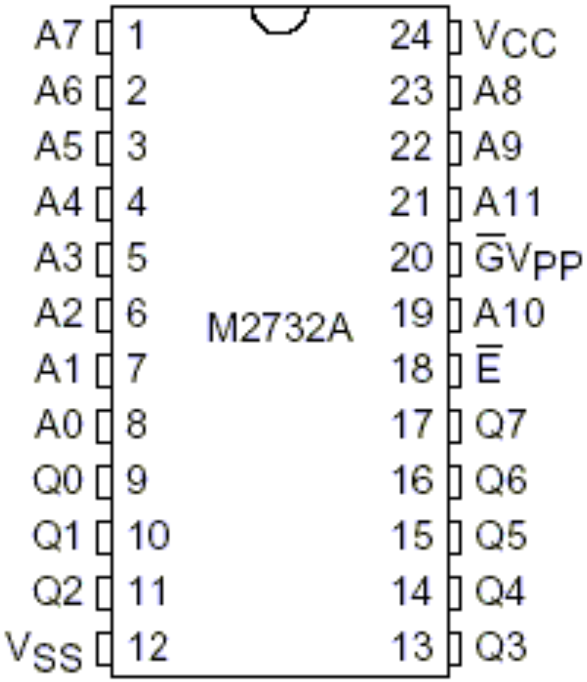




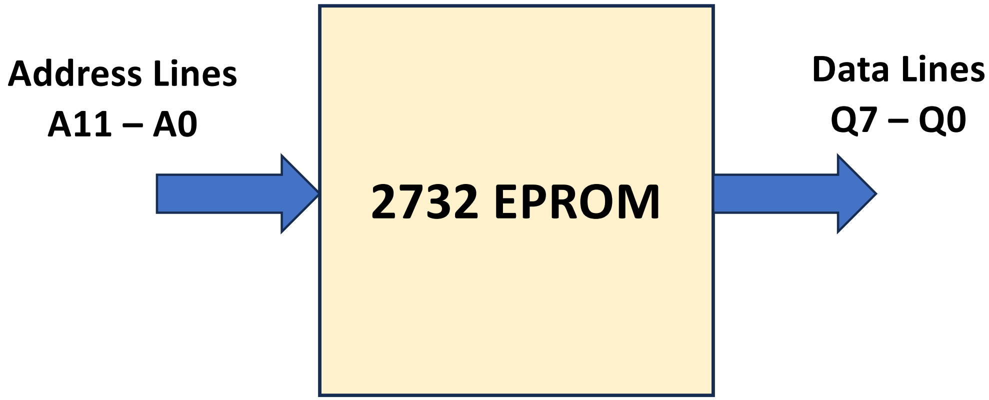
Inputs				Output Y	
Strobe	Select	A	B	DM74LS157	DM74LS158
H	X	X	X	L	H
L	L	L	X	L	H
L	L	H	X	H	L
L	H	X	L	L	H
L	H	X	H	H	L

For Part(c):

- EPROM is a memory that can be erased by exposure to UV light.
- We can store data in the memory using an EPROM programmer.
  - Once programmed, the data remains stored even if power is switched off.
- We can use such EPROM chips to realize arbitrary functions.



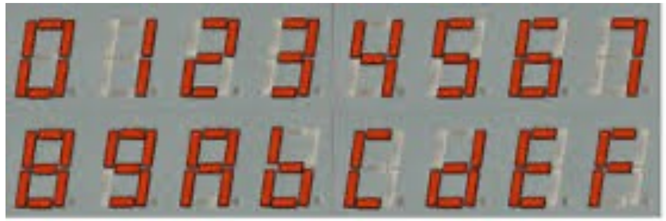
2048 x 8 EPROM

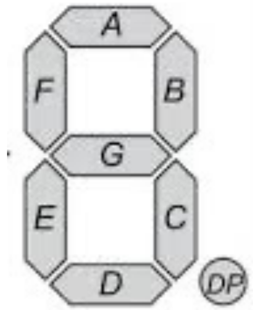


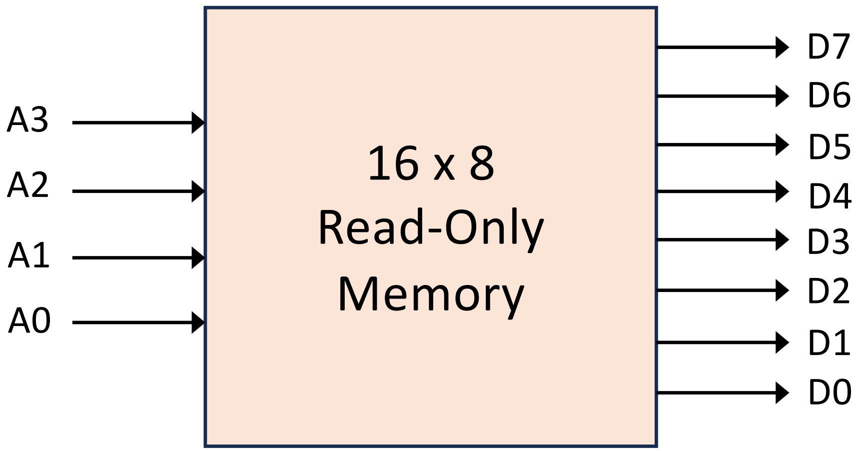
Pin 24 -- +5V
Pin 12, 18, 20 -- GND

4-bit Binary to 7-segment Decoder

b3	b2	b1	b0	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	0	0	0	1
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1
1	0	1	0	1	1	1	0	1	1	1
1	0	1	1	0	0	1	1	1	1	1
1	1	0	0	1	0	0	1	1	1	1
1	1	0	1	0	1	1	1	1	0	1
1	1	1	0	1	0	0	1	1	1	1
1	1	1	1	1	0	0	0	1	1	1







	-	a	b	c	d	e	f	g
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0
11	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0
13	0	0	0	0	0	0	0	0
14	0	0	0	0	0	0	0	0
15	0	0	0	0	0	0	0	0